

Lujain Al-Momani

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Education

University of Wollongong in Dubai, Bachelor of Engineering in Telecommunication & IoT Dubai, UAE

Experience

V2X Networking & Architecture Engineer – Bosch Future Mobility Challenge, Dubai, Sep 2025 – Present UAE

- Led cross-functional development of real-time communication architecture for autonomous systems, enabling data-driven decision making and improving operational accuracy.
- Coordinated multidisciplinary contributors to deliver scalable solutions under constrained timelines using agile execution approaches.

Research Intern – University of Wollongong (UOWD), Dubai, UAE Sept 2024 – Present

- Developed a lightweight machine-learning intrusion detection framework achieving 97% accuracy in constrained environments.
- Conducted analytical evaluation of alternative approaches and translated findings into practical system improvements and published research outputs.

Business Intelligence Intern – Jordan Payments & Clearance Company (JOPACC), July 2023 – Aug 2023 Amman, Jordan

- Produced executive presentations and process manuals supporting adoption of proposed payments system.
- Built foundational skills in Power BI and ETL while assisting with internal reporting.

Publications

- Author of 5 research publications in AI, cybersecurity, and network optimization (ICCE 2025, FICTA 2025, ISDIA 2025; additional papers under review)

Projects

- **AstraIPS: AI-Assisted MQTT Intrusion Prevention System(thesis)** — Built a fog-native, stateful IPS combining Snort 3, a custom MQTT-aware Lua preprocessor, and a BiLSTM model. Developed a custom dataset of benign and malicious MQTT traffic. Achieved 98% accuracy (AUC: 0.9911) with <40 ms latency on edge hardware, with adaptive multi-stage response (alerting to MAC-based isolation). Validated against obfuscated attacks and demonstrated scalability to 80–100 devices. [GitHub Repository](#)
- **Batterly: Smart Pancake Printer** — Led development of an AI-assisted embedded system integrating computer vision and automated calibration to optimize execution efficiency.
- **Autonomous Waste Collection Robot (ERC 2025)** — Built an autonomous robotic system integrating perception and navigation models enabling real-time operational decision making.
- **Hardware-Accelerated Intrusion Detection System for VANETs** — Prototyped a real-time embedded security solution achieving 99.72% detection accuracy while optimizing communication efficiency.

Skills & Certifications

Skills: Structured Problem Solving, Data Analysis & Visualization, Agile Project Management, Stakeholder Collaboration, Process Optimization, Risk Assessment, Strategic Thinking, Team Leadership

Technical Tools: Python, MATLAB, Java, C, Power BI, ROS, Git, VMware, Cisco IOS, Machine Learning, Network Architecture & Security

Certifications: PwC Elevate Me (2025), McKinsey Forward Program (2024), Agile Software Development (Nano

Degree, 2023)

Languages: Arabic, English

Affiliations: IEEE, Engineers Australia

Competitions: HackMyRobot (NYUAD25), ERC (AUS25), Nexus 2.0 (Amity25), YouthxGE23